

System — battery rail, not instrumented

### SoC domain — measured

vcgencmd pmic\_read\_adc, 2 s interval  
12 rails reporting V and I, summed

- cores (VDD\_CORE)
- memory (DDR\_VDD2, DDR\_VDDQ)
- SoC rails (1V1, 1V8, 3V3)

*5,445 samples per run, 3 h*

### excluded

USB camera  
enumerated throughout

Active Cooler  
temperature-driven

Picamera 2 / CSI  
field build only

power board, charging  
third round only

Table 4 reports SoC power and SoC energy per inference.  
EXT5V is logged as a supply-stability check and is not summed; BATT is not logged.  
Battery-rail instrumentation, and therefore system-level energy, remains future work.